

Dexter Fernandes

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AI / Computer Vision Professional

AI and Computer Vision Engineer with 3+ years' experience building image analysis and machine learning systems for industrial operations. Work spans data pipeline development, model training and evaluation across object detection, segmentation and tracking, and edge deployment. Experienced with large-scale computer vision datasets including COCO, Cityscapes, KITTI and PASCAL VOC. Now looking to apply practical computer vision experience in an applied research role focused on infrastructure planning, operational insight and field-level decision-making.

WORK EXPERIENCE

Esbaar • Muscat, Oman

Jan 2023 - Aug 2023

AI Engineer

- Owned the end-to-end computer vision pipeline for real-time perception in harsh industrial environments, covering data curation, model training through deployment and on-site validation, improving F1 by 15% while maintaining deterministic 24/7 operation.
- Built multimodal data ingestion, cleansing, training, and validation pipelines across image, video, and sensor data using Python, Pandas, Airflow, MLflow, DVC, Optuna, and CVAT, processing 3.6 million records and reducing model rollout time from 3 days to under 24 hours.
- Applied image preprocessing and artefact-detection workflows with OpenCV and scikit-image to identify camera-feed contamination, reducing false positives by 13% in production vision pipelines.
- Benchmarked and optimised edge inference workflows across latency, memory, and stability constraints, defining repeatable FP16 validation paths and acceptance thresholds for reliable deployment across hardware targets.
- Led field validation on BP oil-rig deployments, diagnosing model failures under lighting, motion and environmental stressors and translating findings into practical fixes for live operation.
- Analysed model performance and failure patterns from live video deployments, turning F1, false-positive and stability metrics into deployment guardrails and optimisation recommendations for engineering and operations teams.

LivNSense Technologies • Bengaluru, India

May 2020 - Dec 2022

Lead Software Engineer - AI

- Led a team to design, train, and deploy production object detection and tracking models for industrial safety applications, reducing false positives by 33% while meeting latency and power constraints.
- Built low-latency perception pipelines using OpenVINO, and C++17, tripling throughput from 5 FPS to 15 FPS and delivering deterministic real-time inference on Intel VPU hardware under tight power and memory constraints.
- Built automated labelling, dataset versioning, and performance-tracking workflows, repurposing existing detection models for annotation and cutting manual labelling effort by 80% while accelerating iteration cycles on large real-world datasets.
- Conducted optimisation experiments across quantisation-aware training, pruning, layer fusion, FP16/INT8 deployment and profiling, reducing model size by 50% while preserving accuracy under edge constraints.
- Developed synthetic data generation and augmentation tools to improve edge-case coverage and cut manual labelling time by 60% using OpenCV and Albumentations.
- Presented model performance, failure analysis and deployment trade-offs to engineering, QA, hardware and client teams; findings directly shaped rollout and optimisation decisions.
- Used confusion-matrix-driven failure clustering to diagnose recurring CV failure modes, prioritise dataset fixes, and improve model robustness across camera feeds.

PROJECTS

Modern Edge CV Benchmarking

Mar 2026

- Benchmarked modern computer vision models including YOLOv8, YOLOv10, RT-DETR, EfficientViT, Grounding DINO, MobileSAM, and Depth Anything v2 to compare accuracy, latency, model size, and deployment suitability.
- Evaluated Vision Transformer and foundation-model approaches for object detection, segmentation, depth estimation, and open-vocabulary visual understanding, with a focus on production-style CV trade-offs.
- Exported models to ONNX and evaluated OpenVINO deployment paths, strengthening practical understanding of model optimisation, inference acceleration, and edge deployment constraints.
- Built benchmark comparisons using mAP, inference time, and model-size measurements to support model selection for low-latency visual AI systems.

ROS2 SLAM & Nav2 Simulation

Feb 2026

- Built a simulation-based robotics stack in ROS2 Jazzy and Gazebo Harmonic to evaluate LiDAR-centric SLAM, LiDAR+RGB fusion SLAM, and autonomous navigation in structured scenarios.
- Benchmarked RTAB-Map, LIO-SAM, Cartographer, and GLIM across 2D and 3D LiDAR configurations, comparing mapping and localisation behaviour to support system selection and iteration.
- Developed ROS2 nodes, launch files, and parameterised configurations in C++ and Python, using tf2, message synchronisation, camera and sensor ingestion, and timestamp alignment to maintain frame consistency and debug transform timing issues.
- Implemented point cloud processing with PCL and Open3D for filtering, segmentation, and registration, and added an EKF-based localisation layer with robot_localization to fuse wheel odometry and IMU for smoother odom-to-base tracking during LiDAR-only SLAM.
- Containerised the stack with Docker on Linux and managed versioning in Git to support reproducible builds and consistent execution across development environments.

End-to-End Damage Severity & Claim Assessment

Nov 2025

- Developed an automotive claims assessment pipeline combining computer vision-based damage detection with severity classification to support faster triage of real-world vehicle damage scenarios.
- Built a multimodal image-to-structured-output workflow that converted vehicle-damage detections and severity predictions into insurer-friendly claim summaries using structured reasoning and LLM-generated explanations.
- Designed the full workflow across data preprocessing, model training, evaluation, and a user-facing inference module, demonstrating end-to-end delivery from model output to usable product insight.
- Created interpretable repair insights from CV model outputs, translating visual predictions into clear claim explanations for non-technical users.

EDUCATION

Master of Science in Robotics - Merit

University of Bristol • Bristol, England

Sep 2023 - Sep 2024

Dissertation: Radiological SLAM with LiDAR Odometry

- Radiological SLAM with LiDAR Odometry. Designed and implemented a real-time SLAM system using RTAB-Map on the Clearpath Husky platform, integrating LiDAR and radiological data for localisation and mapping in inspection scenarios.
- Evaluated RTAB-Map, LIO-SAM, and Cartographer across comparable scenarios using ROS2 and rosbag2-based workflows to assess mapping and localisation behaviour.
- Validated sensor placement for coverage and calibrated LiDAR extrinsics to the robot base, including the Velodyne VLP-16 and Intel RealSense D435i.
- Implemented a ROS2 (rclcpp) point cloud pipeline in C++17 using PCL, and developed ROS2 nodes with tf2, message synchronisation, and sensor data playback workflows to maintain frame consistency and debug transform timing issues.
- Worked with camera and sensor ingestion in ROS2 pipelines, including timestamp alignment and downstream processing for SLAM and perception tasks.

Bachelor of Technology in Electrical and Electronics Engineering

Visvesvaraya National Institute of Technology • Nagpur, India

Jul 2016 - May 2020

- Built a strong foundation in signals, systems, embedded hardware, and control, which later supported work in real-time AI, sensor systems, and robotics software.
- Completed deeplearning.ai Deep Learning Specialisation alongside the degree, covering neural networks, convolutional models and practical deep learning workflows.

CERTIFICATIONS

Deep Learning Specialisation

Jul 2019

deeplearning.ai

SKILLS

- **Computer Vision & Image Processing:** Anomaly detection, Data augmentation, Defect detection, LiDAR extrinsics calibration, Object detection, Object tracking, Segmentation, Sensor calibration, Visual quality inspection
- **Datasets, Annotation & Evaluation:** Automated labelling, Cityscapes, COCO, Confusion-matrix-driven failure clustering, CVAT, Dataset versioning, F1-score, ImageNet, IoU, KITTI, Latency benchmarking, mAP, Model-size benchmarking, PASCAL VOC, Precision/Recall, Synthetic data generation
- **Deep Learning and Optimisation:** Bayesian Hyperparameter Optimisation, Convolutional Neural Networks, FP16/INT8 deployment, Layer fusion, ONNX, OpenVINO, Optuna, Post-training Quantisation, PyTorch, Quantisation-Aware Training, Scikit-Learn, Structured pruning, TensorFlow, TensorRT, YOLO
- **Computer Vision Models:** ByteTrack, DeepLabv3+, Depth Anything v2, EfficientDet, EfficientViT, Faster R-CNN, Grounding DINO, Mask R-CNN, MobileSAM, RT-DETR, SegFormer, SSD MobileNet V2, U-Net, YOLOv10, YOLOv5, YOLOv8
- **Programming & Tooling:** Bash, C++17, CMake, Git, Linux, NumPy, OpenCV, Pandas, PIL, Python, Scikit-image, Scikit-Learn
- **Cloud, Edge & MLOps:** Apache Airflow, Apache Kafka, API design, AWS, CI/CD, Docker, DVC (Data Version Control), Health checks, Microsoft Azure, MLFlow, Post-deployment validation, Production monitoring, Structured logging
- **Professional Skills:** Agile framework, Experiment-driven development, Stakeholder communication, Technical leadership
- **AI-Assisted Development:** ChatGPT Codex, Claude Code, Prompt engineering